#ManyDaughters: Pre-Analysis Plan

*For **all four hypotheses**, we use the same independent and control variables:*

- Independent variable: we use *kidgeb01 – kidgeb19* (Birth year of up to 19th kid) and *kidsex01 – kidsex19* (sex of up to 19th kid) from the *biobirth* dataset to create our independent variable on having biological daughters or not. Based on variable *syear* and *kidgeb**, we first build a variable on whether the xth kid has been born in the survey year. Relying on this information and information on the sex of the xth kid, we build our independent variable with three categories denoting whether having biological daughters in the survey year: 'No daughter', 'Only daughter(s), and 'Mixed'.

- Control variables: Gender, age group, education, employment status, net household income, marital status, number of biological children, whether youngest biological child <= 3 years, region (West or East Germany), and migration background

  Education, marital status, and employment status are based on variables from the *pgen* dataset; household income based on variables from the *hgen* dataset; number of biological children and youngest child <= 3 years based on variables from the *biobirth* dataset; the rest of the controls are based on variables from the *ppathl* data set.

  We use *sex*, *sampreg*, and *migback* the way they are for measuring the gender, current region (West-East), and migration background of the respondents. We use *syear* and *gebjahr* to build age and then categorise the age variable into four age groups: '<25', '25-39', '40-54', and '55+'. For education, employment status, net household income, and marital status, we use *pgisced11, pgemplst, hghinc, pgfamstd,* respectively, and summarise some of the original categories in each variable. In the end, we obtain three categories for education: 'Low', 'Medium', 'High'; four categories for employment status: 'Full-time employment', 'Regular part-time employment', 'Marginal part-time/(vocational) training', 'Not employed'; five categories for net household income '<1500 EUR', '1500-2499 EUR', '2500-3499 EUR', '3500-4499 EUR', '4500+ EUR'; three categories for marital status: 'Married', 'Single', 'Separated/divorced/widowed'. We build the variable on number of biological children based on the variable we generated earlier on whether the xth kid has been born in the survey year (using variable *syear* and *kidgeb**), with four categories: 'No child', '1 child', '2 children', '3 children or more'. Finally, we use *kidgeb01 – kidgeb19* and *syear* to calculate age of xth child in survey year and then build a dummy on whether the youngest child is 3 years old or younger.

***H1:** Having daughters leads to stronger support for gender equality.*

- Dependent variable: For measuring support for gender equality, we include the following two items from the *pl* dataset: *plh0298_v2*, *plh0302_v2*. On a scale from 1 to 7, respondents were asked how they agree the following three statements, respectively: 1) Children below the age of 6 suffer if their mother works; 2) Children below the age of 3 suffer if their mother works. The two items are highly correlated with each other. We

combine them into one variable by adding them together and reverse the direction of the new generated variable (ranging from 2 to 14): the higher the value, the stronger the support for gender equality.

- Inclusion/exclusion criteria: We only keep participants from private households and focus on heterosexual (based on variable *sexor* in *ppathl* dataset) individuals from 18 to 67 years old. We study the year of 2018 due to the availability of the dependent variables (the first two variables and a similar version of the third variable were also surveyed in 2012 but on a much smaller, completely different group of participants.).
- Exact analysis: We employ multivariate linear regression model (OLS), applying sampling weights and standard errors at the household level. The exact specification of the model is as follows:

$$SupportGenderEquality_{ih} = \beta_0 + \beta_1 OnlyDaughter_{ih} + \beta_2 Mixed_{ih} + \gamma' Controls_{ih} + \varepsilon_{ih}$$

Note: $_{ih}$: Individual i in household h; Our dependent variable (based on items on whether small children suffer when mother works) ranges from 2 to 14, the higher the value, the stronger the support for gender equality; Independent variable has three categories ('No daughter' as the reference category) and we obtain coefficients for 'Only daughter(s)' and 'Mixed', respectively; Control variables are gender, age group, education, employment status, net household income, marital status, number of biological children, whether youngest biological child <= 3 years, region (West or East Germany), and migration background.

- Interpretation of results: We take into account the coefficient estimates and p-value (based on coefficient estimates, standard errors at the household level, residual and model degrees of freedom, and t-test statistics). We interpret the test result as in line with the hypothesis if the coefficient of having daughters (only daughters or mixed) is positive (coef. > 0) and statistically significant (p-value < .05).

H2: *Having daughters leads to more left-leaning political preferences.*

- Dependent variable: We use *plh0004* in the *pl* dataset, measuring political left-right self-placement on a scale from 0 to 10. We reverse the direction of the variable so that higher value denotes more left-leaning preferences.
- Inclusion/exclusion criteria: We only keep participants from private households and focus on heterosexual (based on variable *sexor* in *ppathl* dataset) individuals from 18 to 67 years old. Due to the availability of our dependent variable, we only include the following years: 2005, 2009, 2014, 2019.
- Exact analysis: We employ random effects (RE) linear regression models, applying robust standard errors. The exact specification of the models is as follows:

$$PoliticalPreference_{it} = \beta_0 + \beta_1 OnlyDaughter_{it} + \beta_2 Mixed_{it} + \gamma' Controls_{it} + u_i + \varepsilon_{it}$$

Note: $_{it}$: Individual i at time t, u_i: unobserved individual-specific random effect, ε_{it}: idiosyncratic error term; Our dependent variable: Political self-placement from 0 (right) to 10 (left); Independent variable has three categories ('No daughter' as the reference category) and we obtain a coefficient for 'Only daughter(s)' and for 'Mixed', respectively; Control variables are gender, age group, education, employment status, net household income, marital status, number of biological children, whether youngest biological child <= 3 years, region (West or East Germany), and migration background.

- Interpretation of results: We take into account the coefficient estimates and p-value (for RE model: based on coefficient estimates, robust standard errors, and z-test statistics). We interpret the test result as in line with the hypothesis if the coefficient of having daughters (either only daughters or mixed) is positive (coef. > 0) and statistically significant (p-value < .05).

H3: *Having daughters leads to more gender equality in the household.*

- Dependent variable: We use three variables from the *pl* dataset denoting hours spent during weekdays in childcare, housework, and repair work, respectively (*pli0044_h, pli0043_h, pli0049_h*). We add the hours spent in these three domains together and calculate for each household the male partner's share in domestic work as an indicator for gender equality in the household, ranging from 0 to 100: A higher share of the male partner indicates more gender equality.
- Inclusion/exclusion criteria: We only keep private households and focus on heterosexual (based on variable *sexor* in the *ppathl* dataset) co-habiting couples (based on variable *hgtyp2hh* in the *hgen dataset*) from 18 to 67 years old. We run analysis at the household level and only keep for each household the male partner to account for their socio-demographics. We include all years from 2000 to 2020.
- Exact analysis: We employ fixed effects (FE) linear regression models, applying robust standard errors. The exact specification of the models is as follows:

$$\text{GenderEquality}_{it} = \beta_0 + \beta_1 \text{OnlyDaughter}_{it} + \beta_2 \text{Mixed}_{it} + \gamma' \text{Controls}_{it} + \alpha_i + \varepsilon_{it}$$

Note: $_{it}$: Individual i at time t, α_i: unobserved individual fixed effect, ε_{it}: idiosyncratic error term; Our dependent variable: male partner's share in different domestic work during weekdays (ranging from 0 to 100: the higher the share, the higher the gender equality in the household); Independent variable has three categories ('No daughter' as the reference category) and we obtain a coefficient for 'Only daughter(s)' and for 'Mixed', respectively; Time-varying control variables are age group, education, employment status, net household income, marital status, number of biological children, whether youngest biological child <= 3 years, region (West or East Germany).

- Interpretation of results: We take into account the coefficient estimates and p-value value (for FE model: based on coefficient estimates, robust standard errors, residual and model degrees of freedom, and t-test statistics). We interpret the test result as in line with the hypothesis if the coefficient of having daughters (either only daughters or mixed) is positive (coef. > 0) and statistically significant (p-value < .05).

H4*: Having daughters leads to a larger increase in the support of gender equality for men than for women.*

- Dependent variable: SAME AS FOR H1
- Inclusion/exclusion criteria: SAME AS FOR H1

- Exact analysis: We employ multivariate linear regression model (OLS) with interaction between the independent variable and sex, applying sample weights and standard errors at the household level. The exact specification of the model is as follows:

$$SupportGenderEquality_{ih} = \beta_0 + \beta_1 OnlyDaughter_{ih} + \beta_2 Mixed_{ih} + \beta_3 Male_{ih} + \beta_4 (OnlyDaughter_{ih} \times Male_{ih}) + \beta_5 (Mixed_{ih} \times Male_{ih}) + \gamma' Controls_{ih} + \varepsilon_{ih}$$

Note: $_{ih}$: Individual i in household h; Our dependent variable (based on items on whether small children suffer when mother works) ranges from 2 to 14, the higher the value, the stronger the support for gender equality; Independent variable has three categories ('No daughter' as the reference category) and interacts with gender (whether male or female); Control variables are age group, education, employment status, net household income, marital status, number of biological children, whether youngest biological child <= 3 years, region (West or East Germany), and migration background.

- Interpretation of results: We take into account the coefficient estimates and p-value (based on coefficient estimates, standard errors at the household level, residual and model degrees of freedom, and t-test statistics). For each dependent variable, we interpret the test result as in line with the hypothesis if the interaction term is positive (coef. > 0) and statistically significant (p-value < .05).